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Attractive infrastructure for everyone? Different preferences for route characteristics among cyclists

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ABSTRACT

Knowing about the specific route preferences of demographic groups is crucial when researching equity in cycling. In this study, we use a large-scale stated-preference experiment with 4389 respondents and 35,112 choice observations to evaluate route choice behavior. We set up discrete route choice models and focus on the specific needs of women, people cycling with children, and older adults. Results show that all groups of interest show higher, but not fundamentally different, route requirements. In this context, lower speed limits for motorized transport are very important for people cycling with children, and older adults. In addition, women and those traveling with children attach significantly higher value to separated bike infrastructure. These results show that vulnerable road users benefit stronger from a bike-friendly road infrastructure than the cyclist population as a whole.

1. Introduction

Unlike motorized transport, cycling has major advantages regarding emissions of CO_2, pollutants, and further negative externalities (Makarova, Mavrin, Magdin, Shubenkova, & Boyko, 2019). Cycling as a cost-efficient, healthy mode of transport is also beneficial for the individual (Lee et al., 2012; Mueller et al., 2015). Physical inactivity is known to be a major cause of chronic diseases and significantly harms public health (Blair, 2009; Ding et al., 2016). At the same time, research has demonstrated strong positive interrelations between a better quality and quantity of the bike infrastructure and cycling levels (Buehler & Pucher, 2011; Hardinghaus, Nieland, Lehne, & Weschke, 2021; Winters, Teschke, Brauer, & Fuller, 2016). Consequently, better bike infrastructure is associated with improvements in public health (Mueller et al., 2018). Thus, many cities all over the world are improving cycling conditions and investing in bike infrastructure (Lanzendorf & Busch-Geertsema, 2014).

In recent scientific literature, however, we are experiencing growing concerns regarding the inequity of the developments described. The main question in this discussion is whether all groups in society have the same access to good bike infrastructure and may benefit from the redesign of the streetscape (Sheller, 2015). These equity concerns slowly started to attract more attention when the redesign of the urban road infrastructure picked up speed at the beginning of the 21st century. Before that, in many places the concept of vehicular cycling opposed separated bike infrastructure (Forester, 1993). Accordingly, the group accommodated by such a concept consisted mainly of middle-aged males (Schultheiss, Sanders, & Toole, 2018).

The research body associated with equity of the bike infrastructure is currently growing. Mora et al. (2021) evaluate the accessibility of bike infrastructure in Santiago de Chile. The authors focus on income and conclude that wealthier groups have better access

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to bike infrastructure (Mora, Truffello, & Oyarzún, 2021). Tucker and Manaugh (2018) draw a similar conclusion from a study in Brazil (Tucker & Manaugh, 2018). Similar results are also obtained from multi-city approaches in the US (Braun, Rodriguez, & Gordon-Larsen, 2019; Flanagan, Lachapelle, & El-Geneidy, 2016). Doran et al. (2021) explicitly mentioned women and seniors when claiming that not all groups had access to adequate infrastructure in Canada (Doran, El-Geneidy, & Manaugh, 2021). This research focuses on marginalized groups and income as well as demographic variables. A recent study from New York shows that an upgraded bike infrastructure increases the share of female cyclists (AitBihiOuali & Klingen, 2022).

On the other hand, different route preferences for various sociodemographic indicators are also of interest. There is a large body of research on cyclists' route preferences using various research approaches. Previous studies use stated-preference choice experiments (Clark, Mokhtarian, Circella, & Watkins, 2019; Winters & Teschke, 2010), revealed preference approaches (Broach, Dill, & Gliebe, 2012; Hardinghaus & Nieland, 2021; Pritchard, 2018) or qualitative approaches (Barratt, 2017; Van Cauwenberg et al., 2018). Thus, not all studies focus on investigating different preferences for relevant subgroups or even differentiate between them. When considered, results differ accordingly. In a review of 54 stated-preference studies, Aldred et al. (2016) conclude that women clearly have a stronger preference for separated infrastructure than men do (Aldred, Elliott, Woodcock, & Goodman, 2016). Vedel et al. (2017) find similar results (Vedel, Jacobsen, & Skov-Petersen, 2017). Results also indicate a stronger preference for separated infrastructure among older adults but, due to contradictory findings of the studies included, evidence is limited (Aldred et al., 2016). Other stated-preference studies find little difference between the route choice of women and men (Clark et al., 2019; Rowangould & Tayarani, 2016). Winters and Teschke (2010) investigate route preferences of "the near market for cycling" and find higher route requirements among women as well as people who cycle less often (Winters & Teschke, 2010).

Other research defines the subgroup of interest beforehand and focuses on the specific needs of this group (Garrard, Handy, & Dill, 2012; Ghekiere et al., 2014). In a qualitative approach, Russel et al. (2021) reveal a high burden of women's safety concerns regarding cycling in New Zealand (Russell, Davies, Wild, & Shaw, 2021). Desjardins et al. (2021) in Canada (Desjardins et al., 2021) present another qualitative approach for the case of a city catching up in bike infrastructure. In Belgium, Van Cauwenberg et al. (2018) use bike-along interviews to figure out environmental themes that influence seniors' cycling experience (Van Cauwenberg et al., 2018). Bike-along interviews are also used by Ghekiere et al. (2014) to investigate environmental factors that influence cycling among children (Ghekiere et al., 2014). Such approaches deliver valuable and detailed insights into the specific groups but have limitations when it comes to comparability with the majority of society. Earlier studies also have limitations regarding the sample size, the specific local spatial context, and the integration of comprehensive as well as innovative street properties. As a consequence, it remains unclear to what extent the route preferences of certain groups differ from the majority. This is crucial when aiming to infer recommendations for planning and practice with regard to municipal cycle strategies. In addition, this knowledge enables us to evaluate bike-friendly city development, and in particular its equity conditions, from the perspective of subgroups.

On one hand, equity in the supply of bicycle infrastructure means a spatially equal distribution that all neighborhoods benefit from. On the other hand, it means an infrastructure that is designed in such a way that different groups benefit equally given their specific needs. We can only fully evaluate the equity in supply of bicycle infrastructure if we have better knowledge of these specific needs in all groups of society. This is particularly important, since transport planners are often predominantly middle-aged males which may lead to disadvantages for women in the production of space (Burgess, 2008; Little, 1994; Tummers & Wankiewicz, 2020). Hence, it is crucial to evaluate whether the claims defined and the decisions made meet the demand of groups with less inclusion. At the same time, earlier research was aimed at large-scale quantitative studies to evaluate influences on cycling experience in subgroups (Van Cauwenberg et al., 2018).

Hence, in this study we conduct a large-scale choice experiment right across Germany to investigate route preferences for various relevant road properties. In the evaluation, we focus on the requirements of women, seniors and those cycling with children, since literature suggests that these groups have specific needs regarding infrastructure and corresponding route preferences.

2. Methods and data

In the present study, we use data from a large online survey conducted in 2018 (Hardinghaus & Papantoniou, 2020). In the survey, an online discrete route choice experiment was carried out. It was enriched with sketches presenting the according streetscape to the participants. Respondents had to choose between three different routes with varying travel time and route environments with an opt-out option (choose not to cycle at all). Table 1 shows the attributes of each choice and corresponding levels. In addition to the actual

Table 1
Attributes and levels used in the experiment.

Attribute	Levels
Street type	**Arterial road** | Side street
Cycle infrastructure	**No cycle infrastructure** | Bike lane | Cycle path | Protected bike lane
Regulation	**Maximum car speed: 50 km/h** | Maximum car speed: 30 km/h / Zone 30 |Cycle street (residents only) | Living street
Surface	**Cobblestones** | Asphalt
Parking	**No on-street parking** | On-street parking
Trees	**No trees** | Trees
Travel time [minutes]	8 | 10 | 12 | 15

Bold indicates the reference scenarios. Source (Hardinghaus & Papantoniou, 2020).

experiment, participants were requested to provide various sociodemographic data in the online poll. Survey respondents were recruited via social media, the German national cycle portal, universities, and cycling associations. In total, 4,463 people from Germany participated. In addition, the survey was also translated into other languages and distributed in other European countries. The present study, however, only uses the data from participants in Germany (see section *Sample* for further information on sociodemographic distribution). Due to the survey distribution method, the sample may be self-selective and biased towards regular cyclists, and therefore not representative of the entire population. However, when assessing route preferences of cyclists (regarding cycling infrastructure), it is the respondents who ride a bike that are of particular interest.

The experiment was based on a Bayesian efficient design using the data originating from a pretest designed in a manual design and the software Ngene (Bliemer & Rose, 2006). The experiment was set up with eight choice situations in which respondents had to choose between three route alternatives and the no-choice alternative (meaning the trip would not be taken under the conditions offered) for a recently made trip. For more variance, the choice situations alternated in three blocks (totaling 24 different choice situations). An example choice situation is illustrated in Fig. 1.

Comprehensive information about the conceptualization of the experiment and the recruitment strategy are described in the original resource (Hardinghaus & Papantoniou, 2020).

2.1. Sample

The sample used for the analysis in this paper includes all the observations of respondents from Germany who provided information regarding their age, gender and whether or not they cycle with children. This leads to a total of 4,389 respondents which adds up to 35,112 choice observations from the experiment. As this paper estimates preferences for infrastructure characteristics when riding a bike for different sociodemographic groups, Table 2 shows distributions of age, gender, frequency of riding a bike, and cycling with children for the full sample but also separated according to the user groups analyzed. When compared with national travel survey data, it can be confirmed that there are more male cyclists than female cyclists in Germany, even if the nine percentage points difference is much smaller than in this sample (Nobis, 2019). Compared to the German population, the overall frequency of cycling is much higher in the sample. In Germany, 18% of the population cycle daily, while 36% never or almost never cycle (Nobis, 2019). In the present sample, these groups consist of 75% and 0.8%, respectively. Considering the German population as a whole, cycling levels are highest among teenagers, while in the age groups between 20 and 79 years, the share of regular cyclists (cycling at least once a month) differs by only four percentage points (Nobis, 2019). Hence, young adults are overrepresented in the present sample.

When looking at the sample distributions of the respective user groups, distribution of age, frequency, and cycling with children are pretty similar in males and females. Some differences exist in the 18 to 24-year age group where the share of males is higher than the share of females which is reversed for the 45 to 54 and 55 to 64 year age groups. Also, at 16.5%, the share of women cycling with children is slightly higher than for males at almost 13%. There is a much greater inequality in the distributions differentiated by the factor of whether cycling with children or not particularly with regard to age, which is pretty much to be expected as it is mostly parents with young children who cycle with them. This means that most of the observations for the group with children are among the 25 to 54-year-olds. Distributions of other categories, however, match much more closely. Roughly 2.5% of the respondents are 65 years

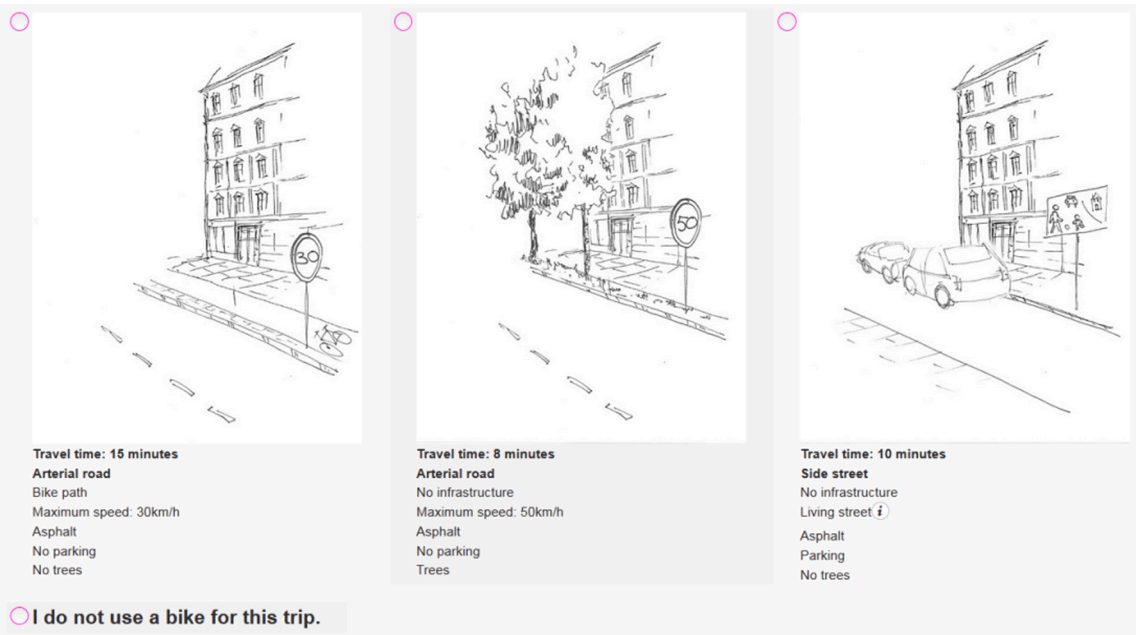

Fig. 1. Example choice situation (Hardinghaus & Papantoniou, 2020).

Table 2
Sample characteristics.

Characteristic	Full sample	Male	Female	Without Children	With Children	Up to 64 years old	Over 64 years old
# Respondents	4,389	3,200	1,189	3,779	610	4,278	111
Age							
Up to 17 years	0.5%	0.4%	0.5%	0.5%	0%	0.5%	–
18 – 24 years	10.3%	11.7%	6.5%	11.9%	0.3%	10.6%	–
25 – 34 years	31.9%	31.5%	33.2%	33.2%	24.3%	32.8%	–
35 – 44 years	23.9%	24.3%	22.7%	19.1%	53.4%	24.5%	–
45 – 54 years	20.9%	19.9%	23.6%	21.1%	19.5%	21.5%	–
55 – 64 years	10%	9.4%	11.4%	11.2%	2.3%	10.2%	–
65 and older	2.5%	2.7%	2.1%	2.9%	0.3%	–	100%
Gender							
Male	72.9%	100%	–	73.8%	67.7%	72.8%	77.5%
Female	27.1%	–	100%	26.3%	32.3%	27.2%	22.5%
Frequency							
Daily	74.9%	74.9%	75.0%	74.5%	77.5%	75.3%	61.3%
1 to 3 days a week	18.7%	18.8%	18.2%	19.2%	15.3%	18.4%	29.7%
1 to 3 days a month	3.9%	3.6%	5.0%	3.8%	5.1%	4.0%	3.6%
Less than once a month	1.6%	1.8%	1.1%	1.6%	1.6%	1.5%	3.6%
Never or almost never	0.8%	0.8%	0.8%	0.9%	0.5%	0.8%	1.8%
No answer	0.1%	0.1%	0%	0.1%	0%	0.1%	0%
Cycling with children							
Yes	13.9%	12.9%	16.6%	–	100%	14.2%	1.8%
No	86.1%	87.1%	83.4%	100%	–	85.8%	98.2%

old or older. The different distributions of sociodemographic variables which exist between these older respondents and the other participants reveal the greatest differences, in contrast to the two previous comparisons. Among the group of older participants, the share of males is even higher (77.5%) than for the other respondents. The older respondents also cycle less on a daily basis and consequently more often only 1 to 3 days a week, which might be due to higher numbers of this group being retirees and resulting in less commuting by bike. Also, at less than 2%, only a very small fraction of older participants cycle with children.

2.2. Modeling strategy

The data from the stated choice experiment is analyzed using multinomial logit (MNL) models. Following McFadden (1973), MNL-models are based on the assumption of random utility theory which says that the utility U of each alternative consists of a deterministic component V and a stochastic component ε (McFadden, 1973). The deterministic component V is the observed part of the utility and depends on the parameter values β. The deterministic utility of alternative i for individual n in a linear functional form can then be described as $V_{ni} = \beta' x_{ni}$ with x_{ni} being the attributes of alternative i for individual n. It is assumed that rational individuals will always choose the alternative which brings them the highest utility. As each respondent of the stated-preference experiment faced eight choice situations, a mixed logit model with error components is applied to account for the panel effect (Train, 2009). In the mixed logit model, the choice probabilities $P_{n,i}$ are the weighted mean of the multinomial logit probabilities $L_{n,i}(\beta)$ over a density distribution $f(\beta|\theta)$ with θ being the parameters of the distribution (mean and variance). The parameters are then estimated using maximum likelihood estimation.

$$P_{n,i} = \int L_{n,i}(\beta) f(\beta|\theta) d\beta$$

$$L_{n,i}(\beta) = \frac{e^{V_{ni}}}{\sum_{j=1}^{J} e^{V_{nj}}}$$

To model differences between the user groups, four models were estimated. Model 1 is the base model, estimating parameters for the infrastructure characteristics as average for all respondents, disregarding their user group (i.e. no interaction terms or other recognition of sociodemographic variables). Models 2, 3 and 4 then estimate the parameters for infrastructure characteristics differentiated by the characteristics gender, cycling with children, and age, respectively, using interaction between the variables. First of all, in each model separate parameters were estimated for all variables and tested for significant difference at 5% level. Subsequently, only those parameters that differed significantly were remained separately in the model, while the others were estimated as joint parameters.

3. Results

The results of the estimated models are shown in Table 3. The estimation results were obtained following the previously explained modeling strategy using the software Biogeme (Bierlaire, 2020) applying 1,000 MLHS draws (Hess, Train, & Polak, 2006). The table is structured such that model 1 shows the parameter estimates for all characteristics disregarding any sociodemographic groups. The

Table 3
Estimation results.

	Model 1	Model 2: Gender	Model 3: Children	Model 4: Age
Attribute	**Estimate (std. error)**	**Estimate (std. error)**	**Estimate (std. error)**	**Estimate (std. error)**
ASC 1	6.84 (0.252)	6.81 (0.244)	6.72 (0.24)	6.77 (0.241)
ASC 2	6.76 (0.252)	6.73 (0.244)	6.64 (0.24)	6.68 (0.241)
ASC 3	6.87 (0.252)	6.83 (0.244)	6.74 (0.24)	6.79 (0.241)
ASC 4 (no choice)	*reference*	*reference*	*reference*	*reference*
Travel time	−0.244 (0.0042)		−0.246 (0.0042)	−0.245 (0.0042)
Travel time male		−0.257 (0.00451)		
Travel time female		−0.211 (0.00603)		
Bike lane	1.04 (0.0322)	1.05 (0.0322)	1.06 (0.0323)	1.04 (0.0322)
Bike path	1.31 (0.0352)			1.31 (0.0352)
Bike path male		1.25 (0.0371)		
Bike path female		1.46 (0.0461)		
Bike path w/ children			1.25 (0.0362)	
Bike path w/o children			1.86 (0.0653)	
Protected bike lane	2.18 (0.0358)	2.18 (0.0359)		2.18 (0.0359)
Protected bike lane w/ children			2.11 (0.0368)	
Protected bike lane w/o children			2.81 (0.0638)	
No infrastructure	*reference*	*Reference*	*reference*	*reference*
Living street	0.664 (0.0259)			0.664 (0.0259)
Living street male		0.611 (0.0282)		
Living street female		0.822 (0.0408)		
Living street w/ children			0.538 (0.0278)	
Living street w/o children			1.45 (0.0705)	
Cycle street	2.1 (0.0391)	2.1 (0.0392)		2.1 (0.0391)
Cycle street w/ children			2.03 (0.0411)	
Cycle street w/o children			2.63 (0.0929)	
Speed limit 30 km/h	0.414 (0.0164)	0.413 (0.164)		
Speed limit 30 km/h w/ children			0.394 (0.0172)	
Speed limit 30 km/h w/o children			0.588 (0.0417)	
Speed limit 30 km/h up to 64y				0.407 (0.0165)
Speed limit 30 km/h over 64y				0.717 (0.0821)
Speed limit 50 km/h	*reference*	*reference*	*reference*	*reference*
Side street	0.68 (0.0327)	0.677 (0.0327)		
Side street w/ children			0.672 (0.034)	
Side street w/o children			0.855 (0.0744)	
Side street up to 64y				0.674 (0.0328)
Side street over 64y				0.979 (0.102)
Main street	*reference*	*reference*	*reference*	*reference*
Asphalt	2.42 (0.0331)	2.43 (0.0331)		2.42 (0.0331)
Asphalt w/ children			2.46 (0.0353)	
Asphalt w/o children			2.34 (0.073)	
Cobblestone	*reference*	*reference*	*reference*	*reference*
On-street parking	−0.681 (0.0139)	−0.683 (0.0139)	−0.686 (0.014)	−0.682 (0.0139)
No parking	*reference*	*reference*	*reference*	*reference*
Trees	0.176 (0.0121)		0.18 (0.0121)	0.176 (0.0121)
Trees male		0.155 (0.014)		
Trees female		0.236 (0.0227)		
No trees	*reference*	*reference*	*reference*	*reference*
Sigma 1	0.142 (0.0547)	0.126 (0.0658)	0.102 (0.0971)	0.129 (0.0625)
Sigma 4	3.42 (0.15)	3.19 (0.145)	3.13 (0.143)	3.16 (0.14)
Respondents	*4,389*	*4,389*	*4,389*	*4,389*
Observations	*35,112*	*35,112*	*35,112*	*35,112*
Log Likelihood	*−34221.92*	−34150.32	−34019	−34086.37
Rho square	*0.297*	0.298	0.301	0.3
LR-Test		143.2	405.84	271.1

following three models then present the results for the estimated parameters separated by either gender, cycling with children or age group. However, when no significant difference at 5% level was found between the two groups, a joint parameter was estimated. The performance of the models is good with rho square values around 0.3 and all parameter signs are as expected without imposing any restrictions on parameters. Testing the model fit of models 2,3 and 4, which consider differences between sociodemographic groups, against model 1 using the likelihood-ratio test, each of the models performs significantly better than model 1. In addition to the variables included in the model, the table includes the reference variable of each group in *italics*, with no parameter estimates to report.

Generally, results show that preferences for some infrastructure characteristics vary between the different sociodemographic groups and in such a way that the specific subgroups und considerations always show the higher preferences. However, not every subgroup values each attribute higher than the average population. Most differences exist between those cycling with children and

those without, while the subgroup of older participants only favor lower speed limits for cars and side streets more. Even when most of the results seem highly plausible, there are some interesting details to note in the differences between males and females. First, females prefer bike paths more than males, but this does not hold true for protected bike lanes, and second, females prefer living streets more than males while this not the case for cycling streets. Since in both cases the infrastructure quality is in ascending order, one may assume that differences in preference also exist for even higher quality infrastructure elements. In these cases, however, it seems that male preferences are the same for the highest quality infrastructure attributes but significantly lower for medium quality attributes.

In the following, some preference comparisons between the groups studied will be pointed out. They always represent the relation between the sociodemographic group under consideration and the outside group (e.g. cycling with children compared to cycling without children). Bike paths are preferred more by females compared to males, and by respondents cycling with children compared to those without. Protected bike lanes (bike lanes separated from motorized traffic by bollards) are favored 33% more among those cycling with children. Preferences for living streets (comparable to shared spaces, cars are only allowed to drive at walking speed) differ between gender and the factor of cycling with children. For the latter group, preferences for living streets are nearly tripled compared to the others while preferences for cycle streets are 30% higher. A speed limit of 30 km/h for cars is especially preferred by those cycling with children and older respondents, with 50% and 75% higher valuation, respectively. This is the same for side streets with around 25% and 50%, respectively, higher preferences compared to the other group. Preferences for asphalt compared to cobblestones are slightly higher for those without children than for those cycling with children. Preferences for on-street parking are the same for all groups analyzed. The presence of trees along the route, however, is favored by females compared to males, with no differences between the other two groups analyzed.

For a better interpretation of the differences in preferences for infrastructure attributes of the respondent groups, the marginal rate of substitution between each attribute and travel time is calculated and presented in Table 4. This means that the values in Table 4 represent the amount of travel time (in minutes) that respondents are willing to invest (on average) in order to obtain the respective infrastructure attribute compared to the reference. For instance, the average population is willing to cycle up to nearly 9 min (8.93) longer to have a protected bike lane on this route. For females, this attribute is worth more than 10 min of extra travel time and those cycling with children are even willing to travel an extra 11.52 min, while those riding without children value the existence of a protected bike lane at only an extra 8.65 min of travel time. In the comparison of valuations in terms of travel time, it is interesting to note that the results of model 2 report a difference in parameter estimates for travel time between women and men, which is why the value of attributes in terms of travel time differs for each attribute even when there is no difference in the parameter for the attribute itself (e.g. bike lane). Due to the difference in parameter estimates, women are always willing to sacrifice more travel time in order to cycle on better infrastructure than men. Differences in valuation of travel time between men and women seem to be in line with previous literature findings on the value of travel time (Schmid et al., 2019). A further advantage of standardizing the parameter values by travel time is the improvement in comparing the valuation between different attributes. While trees are valued nearly twice as high among females compared to males, the overall value is low, so the difference in travel time that they are willing to sacrifice for trees is only half a minute which is much lower than e.g. the value of bike lanes or bike paths.

4. Discussion

The results presented in the four models as shown above are meaningful regarding different preferences among user groups and show that different groups have different demands regarding bike infrastructure. The implications of the results and related recommendations and policy implications will be highlighted in this section.

Comparing the four models, several fundamental trends of different route preferences become obvious. First, all subgroups of interest (women, people cycling with children, and older adults) show higher route requirements than the whole sample. Significantly higher utilities are observed for different route characteristics. There is no case in which the subgroups show lower route requirements.

Table 4
Marginal rate of substitution between each attribute and travel time (in minutes).

Attribute	All users	Male	Female	With children	Without children	Up to 64 years old	Over 64 years old
Bike lane	4.26	4.09	4.98	4.26	4.26		
Bike path	5.37	4.86	6.92	5.12	7.62	5.37	
Protected bike lane	8.93	8.48	10.33	8.65	11.52	8.93	
No infrastructure							
Living street	2.72	2.38	3.90	2.20	5.94	2.72	
Cycle street	8.61	8.17	9.95	8.32	10.78	8.61	
Speed limit 30 km/h	1.70	1.61	1.96	1.61	2.41	1.67	2.94
Speed limit 50 km/h							
Side street	2.79	2.63	3.21	2.75	3.50	2.76	4.01
Main street							
Asphalt	9.92	9.46	11.52	10.08	9.59	9.92	
Cobblestone							
On-street parking	−2.79	−2.66	−3.24	−2.79	−2.79		
No parking							
Trees	0.71	0.60	1.12	0.71	0.71		
No trees							

That means in the trade-off between route characteristics and travel time, the subgroups of interest value the route characteristics analyzed higher than the whole sample does. Second, people cycling with children value slower speeds of motorized transport much higher than the entire population. Riding in a living street is 1.7 times more highly favored by people cycling with children than by those cycling without. To a lesser extent, the preference for lower car speeds can be seen when looking at cycle streets and the speed limit of 30 km/h compared to 50 km/h. The same holds true for side streets which are preferred to a larger extent over main streets. Third, a similar tendency regarding the drawbacks of fast motorized transport is observed in respect of the group of older adults. They value a speed for cars of 30 km/h compared to 50 km/h which is 45% higher than the younger group. For even slower speeds, however, there are no significant differences. Fourth, separated bike-infrastructure (bike paths) is favored more by females than males and by people cycling with children (bike path and protected bike lanes). There are almost no significant differences regarding the road surface, trees along the road and parked cars, factors which can be understood as proxy for the comfort and attractiveness of the road. Only female cyclists value trees along the road higher than males do. A slightly lower preference for a smooth surface within the group of people cycling with children can be explained by generally having permission to cycle on the sidewalk when accompanying a bike-riding child under ten years old.

Compared to earlier research, we find an even stronger preference for separated facilities among women. (Aldred et al., 2016; Vedel et al., 2017; Winters & Teschke, 2010). This finding is in line with current research, indicating that an increase in cycling infrastructure levels the gender balance of cyclists (AitBihiOuali & Klingen, 2022). The authors argue that safety is more important for women than for men. In addition, Grudgings et al. (2021) find that the presence of cycle lanes is more relevant for the share of female cyclists than for males (Grudgings, Hughes, & Hagen-Zanker, 2021). In the present research, however, this finding can only be supported for the case of bike paths. For protected bike lanes, we do not find any significant differences. In relation to other trade-offs, the gender variable is less strong. Hence, overall, we are in line with earlier research which concludes limited differences of route choice regarding gender (Clark et al., 2019; Rowangould & Tayarani, 2016). Regarding the limited evidence of stronger preferences for separated facilities among older people, the present study sees no significant differences (Aldred et al., 2016). Nevertheless, we provide evidence of a stronger preference for side streets and lower speed limits. Recent qualitative research supports calm residential streets and separated bicycle infrastructure as the preferred road environment for cyclists riding their bikes with children (Desjardins et al., 2021). Earlier research suggested that older people had a higher preference for separated infrastructure (Aldred et al., 2016). With regard to this, the present study observes higher route requirements for older cyclists, concluding a stronger preference for calm roads and lower speed limits, but it cannot prove a greater desire for separated infrastructures.

The findings and their consideration in planning and practice are especially important, since older adults and children are vulnerable groups that deserve special protection. Providing a bike-friendly city environment with lower speed zones also makes the city more child-friendly. It allows for independent active mobility and therefore the social participation of minors. At the same time, the socialization of mobility behavior is very important for children, since it influences their behavior over their whole lifetime (Müggenburg, Busch-Geertsema, & Lanzendorf, 2015). Before being allowed to drive, children and teenagers show one of the highest cycling rates across all age groups (Nobis, 2019). A strategy attracting children to cycle even more and ensuring a comfortable and safe experience may therefore have long term effects on the mobility behavior of society. Regarding seniors, earlier research finds that they are more likely to be involved in accidents when cycling (Rodgers, 1997). Furthermore, serious injuries occur more often when accidents are sustained in older age (Dai & Dadashova, 2021; Yang, Yang, Smith, & Robert, 2021). These known higher risks for older adults correspond to their preference for lower speed limits. A good cycle strategy should enable seniors to minimize their individual risk by following their route preferences. Allowing older adults to cycle has a positive influence on their cognitive function and well-being (Leyland et al., 2019).

Based on these findings, recommendations for planning and practice can be drawn. Most important finding of the present study is that all subgroups under consideration show higher route requirements than the mean population. That does not imply that the average population is not valuing proper bike infrastructure, but is less willing to give up travel time (i.e. taking a longer route) to ride on better infrastructure. Consequently, there are no fundamental differences in what an attractive bike-infrastructure looks like. The relevant bikeable road properties are just more important for the subgroups considered. This means that these groups benefit even more than the majority from a bike-friendly adaptation of urban road infrastructure. As a result, slower car speeds are desired to some extent by all subgroups considered. Hence, a strategy of lower speed limits, especially in the lower-level road network, would meet these needs and should therefore be integrated into strategic cycle plans.

The limitations of the present study are discussed in the following. As the data collection approach relied mainly on social media and networks, we expected a self-selective sample of people interested in cycling and corresponding policies. The sample is also distorted in respect of sociodemographic variables as described in section 2. Although the sample size is very large for a choice experiment, the group of older adults accounts for a small share of the whole sample. We do not consider this problematic because the corresponding subsample is still large enough to identify significant differences. Finally, we can only evaluate road properties which were included in the experiment. There may be other attributes or other sociodemographic distinguishing features which would better explain the differences.

5. Conclusion

The specific needs of population groups who are less involved in planning processes are crucial when aiming for equity of supply with bike infrastructure. Results show that the trade-offs between travel time and desired properties of the route differ greatly across the subgroups of interest. It is seen that women, people cycling with children, and older adults have higher requirements regarding bike infrastructure, and in particular place higher value on lower speeds of motorized traffic than middle-aged males. If an alternative

is available, these groups pay for their needs with longer detours and consequently accept longer travel times compared to the average population. Strategies for a bike-friendly city should therefore contain concepts of lower speed limits and generally lower speeds in cities. Results show that specific subgroups benefit more from a bike-friendly redesign than the population as a whole. This applies to each of the user groups under consideration: women, people with children, and older adults.

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CRediT authorship contribution statement

Michael Hardinghaus: Conceptualization, Investigation, Resources, Writing – original draft, Project administration, Funding acquisition. **Jan Weschke:** Writing – original draft, Data curation, Formal analysis, Methodology, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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